

Loan Management Web Application

1. Project Overview

The Loan Management Web Application is designed to help manage individual customer loan details, track repayments, and generate insights through dashboards and AI-based reports. The system will provide a centralized platform for efficient loan tracking and decision-making.

2. Objectives

- Digitize loan management processes
- Provide real-time tracking of loan collections
- Maintain accurate customer records
- Automate repayment updates
- Generate analytical insights using AI

3. Scope

The system will include:

- A secure login system
- A dashboard (homepage)
- Customer management module
- Loan collection module
- Backend database with real-time updates

4. User Roles

- **Admin / Owner**
 - Manage customers
 - Add loans and repayments
 - View reports and dashboard

5. Functional Requirements

5.1 Authentication Module

- Secure login system (username & password)
- Session management
- Logout functionality

5.2 Homepage (Dashboard)

The homepage will provide an overview of the system:

Features:

- Total loan amount given
- Total amount collected
- Pending loan amount
- Number of active customers
- Recent transactions
- AI-based insights (e.g., risk prediction, repayment trends)

5.3 Customer Management Page

5.3.1 Add New Customer

Fields required:

- Customer Name
- Contact Details
- Address
- Loan Amount
- Interest Rate (%)
- Loan Duration (Time Period)
- Monthly Settlement Amount (Auto-calculated based on interest)

5.3.2 Existing Customer List

- Display all customers
- Search and filter functionality
- Click to view full details

5.3.3 Customer Details View

- Customer basic information
- Loan details
- Payment status
- Remaining balance
- Full transaction history (all previous payments)

5.4 Collection Page (Installment Management)

Features:

- Search customer by name
- Add installment payment:
 - Payment date
 - Amount paid
- Automatic updates:
 - Update customer balance
 - Reflect changes in dashboard
 - Store transaction in history

6. Data Flow & System Behavior

- When a new loan is created → stored in database
- When installment is added:
 - Customer balance is updated
 - Transaction is recorded
 - Dashboard metrics are updated in real-time

7. Non-Functional Requirements

- **Performance:** Fast data retrieval and updates
- **Security:** Encrypted login and secure data storage
- **Scalability:** Ability to handle increasing customers
- **Usability:** Simple and user-friendly UI

- **Reliability:** Accurate calculations and data consistency

8. Assumptions

- Only authorized users can access the system
- All loan calculations are based on the Inputs
- Internet connectivity is available